

Jin Woo Oh

ojw925@gmail.com

201-753-0265

Atlanta, GA 30340

[linkedin.com/in/jinwooh](https://www.linkedin.com/in/jinwooh)

[ojw92.github.io](https://github.com/ojw92)

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, HTML/CSS, C++ (object-oriented)

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, LangChain, LLM (Claude, GPT), generative AI (GenAI), prompt engineering, fine-tuning, RAG, MLOps

Cloud & DevOps: AWS, GCP, Kubernetes, Docker, CI/CD (GitHub Actions), Git, monitoring & alerting, agile development

Data & APIs: PostgreSQL, NoSQL, REST API, microservices, FastAPI, unit/integration testing (pytest, Playwright)

EXPERIENCE

Avnir **Nov 2025 – Present**
Software Engineer *Atlanta, GA*

- Diagnosed 45-minute production LLM pipeline jam via root cause analysis; added KEDA autoscaling, queue-backlog monitoring, and alerting to prevent recurrence
- Migrated 3 production GenAI microservices (vLLM inference servers) from Qwen2.5 to Gemma4 on Modal, cutting compute costs by 59% and eliminating cold-start failures
- Built LLM-powered contact-extraction pipeline using Claude prompt engineering and structured JSON output, reducing ingestion time from 40 to 4 minutes
- Engineered relationship-strength scoring model with 30 behavioral features using scikit-learn grid search and cross-validation, cutting pipeline reruns from 75 minutes to 5 seconds
- Shipped 30-chart Apache Superset KPI dashboard through CI/CD across 3 environments, automating Customer Success reporting and saving 12 hours weekly
- Developed GeoPin location-intelligence microservice end-to-end: Python REST API with OAuth authentication and React/TypeScript UI, serving graph-based recommendations from PostgreSQL, exposed over MCP for in-house AI agents

ICC Lab, Florida State University **May 2025 – Nov 2025**
Researcher *Tallahassee, FL (Remote)*

- Automated 1,000+ hours of manual labeling by building a gaze-tracking computer-vision pipeline with OpenCV, overlaying real-time gaze regions on screen recordings
- Achieved 95% accuracy in detecting UI state changes via robust template matching and HSV color masking

FlightAware **June 2024 – Aug 2024**
Software Engineer Intern *Houston, TX (Remote)*

- Reduced deep-learning web application latency by 35% by developing a scalable, secure FastAPI microservice on Kubernetes for internal access across teams
- Evaluated GPT-4o vs Mistral 7B RAG chatbots through A/B testing; recommended GPT-4o for 21x speed and 40% higher success rate
- Improved website reliability by 20% by streamlining REST API integration tests in Postman and reconfiguring Redis and Triton inference-server dependencies

Samsung Electronics America **Aug 2019 – Apr 2024**
Software Quality Engineer III *Berkeley Heights, NJ*

- Spearheaded GPU-accelerated NLP classifier detecting critical flagship-device complaints at 92% accuracy, automating 11+ hours of manual triage weekly
- Built automated ETL pipeline (Python & SQL) processing 1M+ articles for daily QA reports, accelerating issue detection by 25%
- Reduced tech-support live-chat operating costs by 30% via discrete-event simulation, optimizing staffing allocation

PROJECTS

[Technical Issue Classifier](#) | PyTorch, HuggingFace, Docker, Kubernetes

- Optimized classifier decision threshold via ROC analysis and grid search, reducing costly false negatives to 3%

[RayTracing](#) | C++, CUDA, GPU acceleration

- Implemented CUDA ray tracer achieving 35x GPU speedup over CPU via parallel programming and HPC scaling

EDUCATION

Georgia Institute of Technology **Expected Dec 2026**
Master of Science (M.S.) in Analytics *Atlanta, GA (Online)*

Seoul National University **Feb 2018**
Master of Science (M.S.) in Mechanical Engineering *Seoul, Korea*

- 2 patents, 2 journal publications, 8 conference presentations; Academic Scholarship; 3.75/4.0 GPA, *Summa cum laude*

University of Illinois at Urbana-Champaign **May 2015**
Bachelor of Science (B.S.) in Mechanical Engineering *Champaign, IL*

- 1 journal publication; VP, Co-founder, mentor and tutor at Korean Mechanical Engineer Network; Academic Scholarship